

TRANSGlobe_Business

BUSINESS SQL ANALYSIS

Stakeholder-driven analytical requirements for logistics operations

Business	TransGlobe Logistics Pvt. Ltd.
Industry	Third-party logistics (3PL) & last-mile delivery
Network	25 major Indian cities · 10 carrier partners · 30 warehouses
Customer Base	3,000+ customers
Shipment Volume	22,000+ shipments analyzed

Business Context

The following SQL analysis contains a set of **business-driven analytical requirements provided by stakeholders** to evaluate TransGlobe's shipment and logistics operations.

The requirements reflect a real-world analytics workflow: business and operations teams define the questions they need answered, and the Data Analytics team uses SQL to extract, aggregate, compare, rank, and interpret shipment data from the company's data warehouse.

Why the Analysis Is Required

TransGlobe has experienced increasing customer complaints and SLA penalty claims related to late deliveries. The existing delay-diagnosis process is largely reactive, with analysts investigating carrier, weather, warehouse, and route factors after a shipment has already been delayed. The business therefore needs structured analysis to identify performance patterns, operational risks, and improvement opportunities.

Business Objective

Transform stakeholder requirements into actionable SQL-based insights that support **operational decision-making, delivery-performance monitoring, cost optimization, carrier and route evaluation, warehouse efficiency, and proactive late-delivery risk management.**

Data Model

The accompanying schema documentation describes the warehouse as a **Snowflake schema**: Fact_Shipments is surrounded by dimensions, with Dim_Locations shared as an outrigger and Dim_Vehicles / Dim_Employees further normalized. Fact_Shipments has one row per shipment.

Documented tables include Fact_Shipments, Dim_Dates, Dim_Customers, Dim_Products, Dim_Employees, Dim_Vehicles, Dim_Routes, Dim_Warehouses, Dim_Locations, and Dim_Carriers.

Analytical Requirements

The 40 requirements below represent business requests for analysis. They are organized by business area for readability rather than by SQL difficulty.

Core Shipment & Delivery Performance

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| Q1 | How many total shipments are in the dataset? |
| Q2 | What is the overall late-delivery rate across all shipments? |
| Q3 | How many shipments fall into each ShipmentStatus? |

Q4	What is the average shipping cost across all shipments (nulls excluded)?
Q5	What is total order value broken down by shipping priority?
Q6	How many distinct customers have shipped at least once?
Q7	How many shipments were made per delivery mode?
Q8	How does average promised delivery time compare to average actual delivery time?
Q9	How many shipments were made per payment method, most used first?
Q10	What are the min, max, and average shipment distance, and how many rows have invalid (negative) distance?

Carrier, Customer & Operational Analysis

Q11	What is the late-delivery rate for each carrier, worst first?
Q12	What is the late-delivery rate under each weather condition?
Q13	What is the average shipping cost by customer region?
Q14	How does late-delivery rate vary across distance buckets (intra-city to long-haul)?
Q15	Which 5 cities generate the highest total order value?
Q16	How does late-delivery rate vary by shipping priority AND carrier type together?
Q17	What are average handling and loading times by warehouse type?
Q18	Which customers have placed more than 10 shipments, and what's their average prior-delay count?
Q19	After normalizing inconsistent text casing, what is the late rate by traffic condition?
Q20	Do fragile products cost more to ship and run later than non-fragile ones?

Trend, Ranking & Cost Analysis

Q21	Rank every carrier by late-delivery rate.
Q22	What was each customer's most recent shipment?
Q23	How does daily shipment volume change day-over-day?
Q24	Split shipments into shipping-cost quartiles — does late rate rise with cost?
Q25	What is the cumulative running total of order value over time?
Q26	How is the month-over-month late-delivery rate trending?
Q27	For each route, which carrier handles the most shipments?
Q28	Which shipments cost more than their own carrier's average shipping cost? (correlated subquery)
Q29	What percentage of total order value does each product category contribute?
Q30	Rank Traffic x Weather combinations by late-delivery risk.

Advanced Management & Risk Analysis

Q31	Build a carrier scorecard combining late-rate rank and cost rank into one weighted score.
Q32	Cohort analysis: registration-month cohorts vs. their shipment late rate & spend.
Q33	Which carriers have a late rate more than 1 standard deviation above the network average?
Q34	Which routes' late rate worsened by more than 5 points quarter-over-quarter?
Q35	For each warehouse, who is the most time-efficient employee (min. 30 shipments handled)?
Q36	Build a Traffic x Weather late-rate matrix (weather conditions pivoted into columns).
Q37	Flag customers whose latest carrier runs above-average late rates AND whose delay count is rising.
Q38	Do above-median-age vehicles run a higher late rate than at/below-median-age vehicles?

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| Q39 | Top 5 products per category by revenue, with rank and cumulative % of category revenue. |
| Q40 | Flag anomalous shipping costs (z-score > 3) within their own distance-decile peer group. |
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Data & Analytics Notes

The schema documentation reports approximately 22,110 shipment rows, 1,096 dates, 3,000 customers, 800 products, 900 employees, 500 vehicles, 350 routes, 30 warehouses, 25 locations, and 10 carriers — approximately 29,821 rows and 87 documented columns in total.

For the downstream machine-learning use case, ActualDeliveryDays and DeliveryTimeHours are target-leakage variables for predicting Late_Delivery. They can support descriptive BI analysis, but should not be used as predictive features.

TransGlobe Logistics Pvt. Ltd. is a fictional company created for the portfolio project; the underlying dataset is synthetically generated and does not represent a real organization.